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REVIEW

Efforts to control *Lygodium microphyllum* in Florida and developments in biology and management

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Abstract

Lygodium microphyllum, Old-world climbing fern, invades remote and intact natural areas of marsh, wetlands, and swamps within Florida's Greater Everglades Ecosystem. Despite extensive efforts, *L. microphyllum* continues to evade control throughout its introduced range. Old-world climbing fern was introduced to Florida through the ornamental horticulture trade in the early 20th century. In the late 1960s, land managers called attention to its escape from cultivation and its rapid establishment and expansion into natural conservation areas. During the next decades, Old-world climbing fern expanded its range into remote parts of Everglades National Park Wilderness and into areas north of Jacksonville, Florida. This tropical climbing fern smothers native vegetation, alters carbon cycles and fire regimes, and spreads through airborne, self-compatible spores. Treatment using physical removal methods is damaging and labor intensive, chemicals translocate poorly, and thus far, biological control has not effectively limited the spread of this species. All these methods though, continue to be refined and integrated through extensive collaboration with researchers and land managers. Herein we describe the current state of the research programs aimed at finding effective and sustainable control for this invasive plant, and those elucidating the biological and ecological mechanisms behind its invasion.

Keywords: old-world climbing fern, biological control of weeds, Florida, USA, integrated management

Introduction

Old-world climbing fern, *Lygodium microphyllum* Cav. R. Br. (Pteridophyta: Lygodiaceae), is a cosmopolitan species of climbing fern that threatens North America's largest wetland ecosystem, the Everglades (Volin *et al*., 2004; Hutchinson *et al*., 2006). *L. microphyllum* was introduced into the United States through the horticulture trade (Hutchinson and Langeland, 2010; Hutchinson and Hutchinson, 2013; Pemberton and Ferriter, 1998). Plants labeled as *L. microphyllum* were sold in nurseries in Palm Beach County and Hardee County, both of which were early sites of naturalization (Beckner, 1968). In the 1960s land managers reported naturalized populations in Martin County, which spread to Palm Beach and Broward Counties by the 1980s and Collier and Seminole Counties by the 1990s (Pemberton and Ferriter, 1998). By the early 2000s, *L. microphyllum* had colonized much of South Florida on both coasts and additional naturalized populations were observed further north in Orlando and Jacksonville, Florida (30.3322°N, 81.6557°W) (iNaturalist. Available at: https://www.inaturalist.org, accessed 22 February 2025) (Fig. 1).

Review methodology

We utilized the search engines Google Scholar and Web of Science to accumulate references pertaining to *L. microphyllum*. We also searched PhD and Master's degree theses for pertinent results, especially for known collaborators. Finally, we spoke to people with extensive knowledge of the system and, when needed, used these as personal communications.

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Environmental impacts

The relatively rapid establishment and expansion of *L. microphyllum* brought about extensive negative impacts (Gordon, 1998). The climbing and twining habit that *L. microphyllum* exhibits causes mechanical damage to vegetation, and acts as a fire ladder, carrying low-intensity ground fires into canopies that would normally withstand minor fire damage (Hutchinson *et al*., 2006; David *et al*., 2020) (Fig. 2). In sensitive sawgrass marshes (i.e., dominated by *Cladium mariscus* (L.) Pohl ssp. *jamaicense* (Crantz) Kük. [Cyperaceae]), *L. microphyllum* covers large areas in remote and inaccessible parts of Everglades National Park, a federally designated Wilderness (Volin *et al*., 2004). Within the Greater Everglades Ecosystem, *L. microphyllum* invades Everglades tree islands (dominated by tropical hardwood species such as *Swietenia mahagoni* (L.) Jacq. [Meliaceae]), cypress domes (dominated by *Taxodium distichum* L. [Cupressaceae]), and inland mesic hardwood forests and pinelands (dominated by *Acer rubrum* L. [Sapindaceae] and *Pinus elliottii* Engelm [Pinaceae]) (Brandt and Black, 2001; Pemberton *et al*., 2002). Finally, south Florida is prone to severe impacts from the Atlantic hurricane season. In a post-hurricane study, Lynch *et al.* (2009) found that *L. microphyllum* rapidly proliferated in treefall zones, halting recovery of the native vegetation.

Fire is a frequent natural phenomenon in Everglades habitats, including those invaded by *L. microphyllum* (Gunderson and Snyder, 1997; Lockwood *et al*., 2003). Fires in the Everglades, which are predominately started by lightning, often act to reset communities back to an early successional stage (Gunderson and Snyder, 1997).



Fig. 1. The worldwide range of *Lygodium microphyllum* extending from its native range in Australia, Asia, and Africa to its invasive range in Florida (inset). Data points are observation points from iNaturalist. Available at: https://www.inaturalist.org (accessed 22 February 2025).

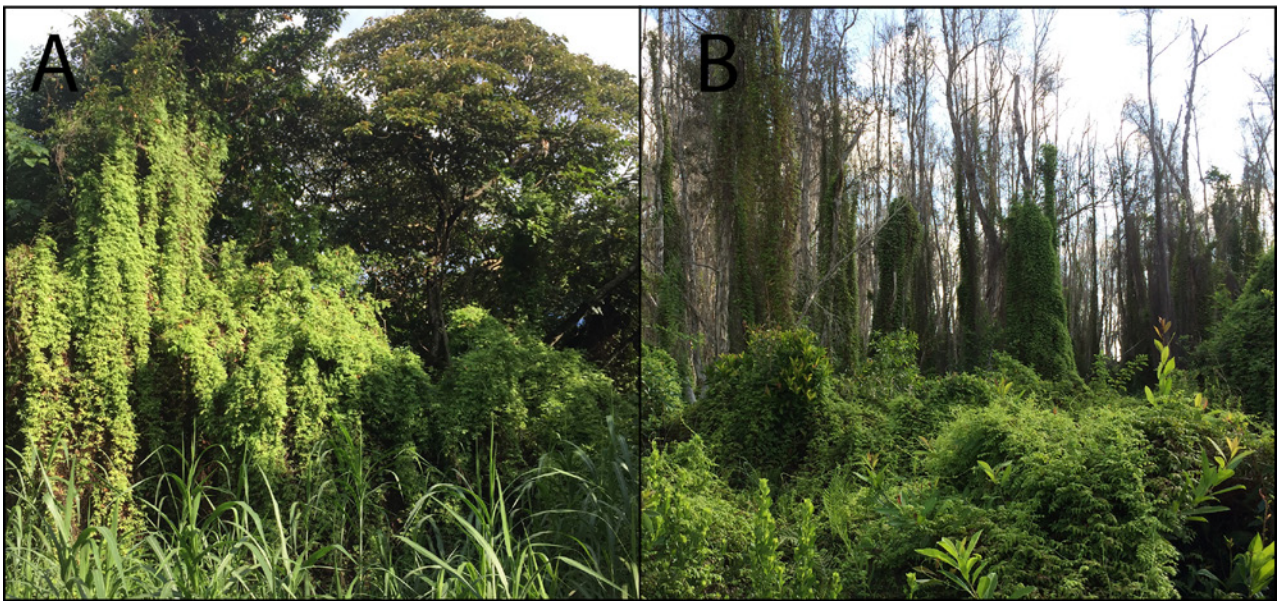


Fig. 2. *Lygodium microphyllum* (A) in its native range in Hong Kong (photo credit: M.C. Smith) and (B) in the Arthur R. Marshall Loxahatchee National Wildlife Refuge (photo credit: E.C. Lake).

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Under natural conditions, sawgrass communities return within 2 years post-fire, and other communities also follow a predictable post-fire succession (Gunderson and Snyder, 1997; Richards *et al*., 2020). When *L. microphyllum* invades these communities, it adds fuel and structure; as a result, fire reaches hotter temperatures and higher into the canopy (Sebesta, 2021). For example, most mature trees survive a ground fire between 2 and 3 m in height, but cannot survive a fire that reaches 20 m into the canopy using dead *L. microphyllum* rachis as a ladder (Roberts, 1996).

The Lygodiaceae is somewhat unique among ferns – it is one of the only families whose rachis grows indeterminately and twines up canopies and over understory vegetation (Garrison Hanks, 1998; Pelosi *et al*., 2024) (Fig. 2). *L. microphyllum* spreads vegetatively and also reproduces sexually through specialized spore-forming pinnae (Lott *et al*., 2003) (Fig. 3A). Spores germinate and reproduce via sexual recombination (i.e., outcrossing or sporophytic selfing), but can also capitalize on an extreme form of inbreeding (gametophytic selfing): a single spore can self-fertilize and create a new sporophyte (Lott *et al*., 2003; Volin *et al*., 2004). Lott *et al*. (2003) showed that *L. microphyllum* has a selfing rate of over 75% in Florida.

Once a *L. microphyllum* spore lands, its germination and establishment are often dictated by a suite of environmental factors, including temperature, light, humidity, salinity and soil mycorrhizae (Hutchinson and Langeland, 2010, 2014; Soti *et al*., 2014; Sebesta *et al*., 2016; Sebesta, 2021). Under dry conditions, spores may remain viable for up to 7 years, a stark contrast to native ferns, including the North American native congener, *L. palmatum* (Bernh.) Sw., which only displays viability for approximately a year (Hutchinson and Langeland, 2010). Call *et al*. (2007) demonstrated that *L. microphyllum* successfully germinate on a variety of different substrates, including soil, leaf litter and water. Gametophytes are hygrophytic and are often seen on the sides of cypress, riverbanks or other areas with persistent moist substrate (Pemberton and Ferriter, 1998). In a series of freezing tolerance experiments, Hutchinson and Langeland (2014) showed that the gametophyte stage is extremely vulnerable to any freezing and is rapidly killed, though recent data suggest that the gametophyte is more tolerant to freezing than the leaf (Pelosi *et al*., 2025). Established rhizomes withstand freezing for longer than 6 h, after which they readily resprout (Pemberton, 2003; Hutchinson and Langeland, 2014). In a greenhouse experiment, salinity affected both sporophyte growth and spore germination (Sebesta, 2021). Spores exhibited delayed germination in 5 and 10 ppt, while in 15 ppt, germination was completely prevented. Sporophytes in 0 and 5 ppt exhibited growth rates similar to those documented by others, but plants treated with 10 ppt saline solution showed signs of salt stress within a few weeks, with several plants dying within 11 weeks (David and Lake, 2020; Volin *et al*., 2010). Salinity stress was associated with a shift to resprouting leaves from the rhizome as – a mitigation strategy seen in plants with neither the ability to exclude salt at the roots nor the capacity for compartmentalization (Munns and Termaat, 1986).

Soil mycorrhizal associations in the new range may increase the ability of *L. microphyllum* to flourish in otherwise unsuitable habitats, and may set off a feedback loop that favors *L. microphyllum* (Soti *et al*., 2014). In work done in growth chambers with soil and soil flora samples from the native range in Australia and the invaded range in Florida, Soti *et al*. (2014) found that *L. microphyllum* formed extensive mycorrhizal associations in dry soils from Florida. While not a direct test, it suggests that *L. microphyllum* is able to proliferate in dryer sites in Florida due to mycorrhizal-mediated phosphorus and water uptake (Soti *et al*., 2014).

While environmental factors such as temperature, humidity, and light attenuation play a role in the germination and establishment of *L. microphyllum*, the interactive effects of these factors and their implications for control strategies remains an area for further exploration.

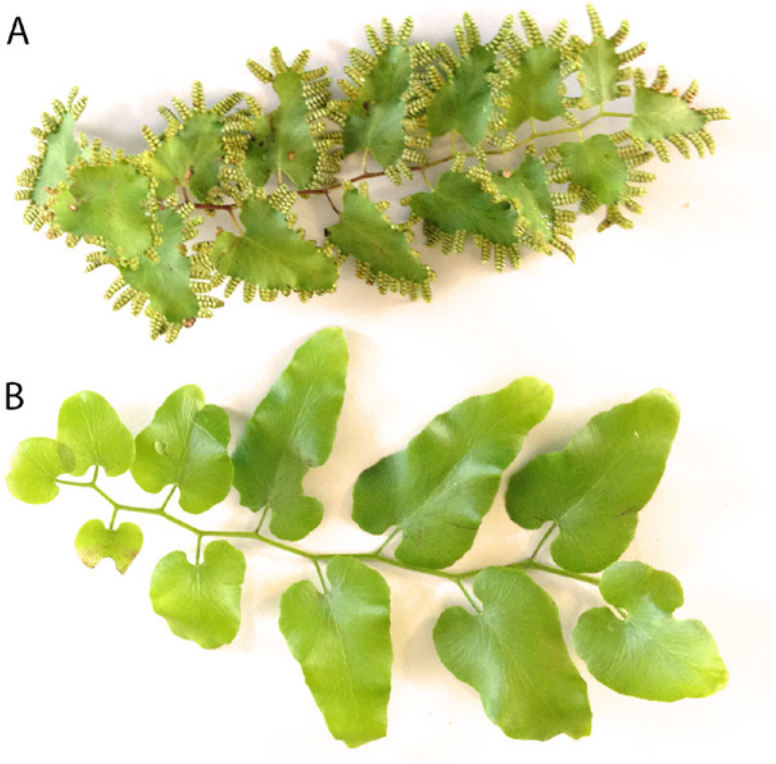


Fig. 3. *Lygodium microphyllum* pinnae types. The fertile (A) pinnae bear spores on modified leaves with sori on the margins. The sterile (B) pinnae display only entire margins.

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Current methods of control

CHEMICAL

Herbicides are currently the most common method to control *L. microphyllum* infestations and are frequently applied as part of an integrated pest management (IPM) strategy (Hutchinson *et al*., 2006). For *L. microphyllum*, herbicide treatments have the potential to influence each of its life history stages, including spores, gametophytes, and sporophytes. IPM, in part, seeks to identify the weak links in the life cycle of the target plant and apply management to exploit that weakness (Norris *et al*., 2003). For this species, the weakest link is still unclear. The charismatic and impactful nature of the sporophyte stage has warranted the most attention from a research and management perspective as most efforts have focused on the control of sporophytic growth. Gametophytes and spores are microscopic in size, largely requiring magnification to even observe, much less study. This has resulted in few studies that have examined the impact of herbicides on spores or gametophytes.

Herbicide treatment of *L. microphyllum* is difficult and often complicated by the remoteness of wetlands with variable hydrology, the broad array of desirable native plants found in invaded wetlands, and high treatment costs (Hutchinson *et al*., 2006). Variable seasonal hydrology frequently limits herbicide treatment. For example, high water during the summer wet season may inundate infested areas, slowing ground crews and making foliar treatment of the submersed rachis mat impossible (Taylor, 2006). Treatment logistics in the wet season also become difficult with high threat of severe weather (Duever *et al*., 1994). Low water during the dry season also limits access to remote infestations. In marshes with shallow water, seasonal fluctuations of even 20–30 cm in water depth shift vast areas of open water to impassable mud flats where neither airboat nor swamp buggy can access (Brandt *et al*., 2003). This is typical in the Arthur R. Marshall Loxahatchee National Wildlife Refuge (LNWR), where aerial surveys identified 74% of the *L. microphyllum* in the Everglades watershed occurs (Rodgers *et al*., 2014).

When access is feasible, applicators face frequent non-target issues. Wetlands are home to a rich array of woody and herbaceous plants that share an intimate association with *L. microphyllum* (Loveless, 1959; Brandt and Black, 2001). *L. microphyllum* overtops multiple vegetation layers within wetlands making directed herbicide sprays extremely difficult (Fig. 4A). The practice of cutting all climbing vines



Fig. 4. Everglades tree islands (A) in ARM Loxahatchee National Wildlife Refuge. Ground crews arrive to the site via airboat to prepare to apply herbicide, (B) Example of poodle cutting technique where *L. microphyllum* is cut at chest height to separate vertical growth into trees from the rhizome, (C) An applicator sprays herbicide mixed with blue dye on *L. microphyllum* in mixed vegetation with desirable native plants (photo credits: J. Glueckert, S. Enloe).

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prior to treatment (i.e., "poodle cutting") can greatly alleviate non target damage to shrubs and trees (Thomas Jr and Brandt, 2003; Barrett *et al*., 2006)(Fig. 4B). However, co-occurring native species within the herbaceous layer often result in non-target damages to desirable native plants (Hutchinson and Langeland, 2012) (Fig. 4C). This is exacerbated by a lack of selective herbicides to target *L. microphyllum* (expanded below).

Finally, *L. microphyllum* management with herbicides is economically costly. *L. microphyllum* ranks in the top five plant species managed on public conservation lands in Florida, with expenditures of over US $2 million annually (Hiatt *et al*., 2019). Private landowners may also apply herbicide treatments to private lands or hire contractors. However, treatment cost is often a limiting factor on private lands and many landowners are unable to treat without federal or state assistance programs. This frequently results in many private lands unintentionally serving as propagule sources of reinfestation (Hutchinson and Langeland, 2006). Ground treatments often range from US$$300 to 1100 per acre depending on site conditions (Hutchinson *et al*., 2006). Aerial applications are generally cheaper, averaging around US$100 per acre (Hutchinson *et al*., 2006).

Foliar treatment using ground-based and aerial approaches is the primary method of *L. microphyllum* herbicide treatments. Helicopters apply herbicide for large infestations. Unmanned aerial systems or UAS-based application is gaining interest for treatment of both large patches and small, incipient infestations. Aerial treatments are generally only conducted in remote areas where dense *L. microphyllum* cover has topped the canopy and collapsed the native woody vegetation. In these cases, the risk of non-target damage is lessened as *L. microphyllum* increasingly dominates.

Ground-based motorized approaches include high volume handgun sprayers mounted on trucks, all-terrain vehicles, amphibious vehicles, and airboats. These are commonly used where infestations are not access limited, such as along roadsides, rights of ways, rivers or canals. These often utilize high application volumes but can be very efficient for treatment of small patches, especially along linear corridors. Ground-based treatments may also include backpack spray crews which are widely used for small, patchy infestations where motorized access is limited (Hutchinson and Langeland, 2006). Ground crew treatments typically involve an integrated cut and foliar spray technique ("poodle cutting"), where crews cut the climbing rachis of the fern at approximately 90 cm and the follow up with a foliar herbicide treatment to all living vines below the cut (Thomas Jr and Brandt, 2003; Barrett *et al*., 2006) (Fig. 4B). Without connection to the soil, the severed climbing vines die without additional herbicide treatment. This targeted approach can greatly increase selectivity, minimize damage to non-target vegetation and reduce applicator exposure during treatment (Hutchinson *et al*., 2006).

In general, since *L. microphyllum* is primarily managed in wetlands where standing water is perennially present, herbicides labeled in aquatic settings are used in most cases. Prior to labeling, aquatic herbicides require additional testing to ensure safety in the aquatic environment to minimize risks to potable water sources, recreation (e.g., swimming and fishing), and irrigation (e.g., agricultural crops or turfgrasses and ornamental plants) (Getsinger *et al*., 2011).

Early studies identified glyphosate, triclopyr, and metsulfuron as effective active ingredients to suppress *L. microphyllum* (Stocker *et al*., 1997; Hutchinson *et al*., 2006; Hutchinson and Langeland, 2007). These continue to be the most used herbicides (Hutchinson and Langeland, 2006) (Table 1). Glyphosate, a systemic non-selective organophosphate, has been the primary tool for *L. microphyllum* treatment (Hutchinson *et al*., 2006). Following treatment, symptomology may appear slowly, over a period of 1to 4 weeks for many herbaceous perennials such as *L. microphyllum*. Common symptoms include a stoppage in growth, foliar chlorosis (yellowing of the leaves), and subsequent necrosis (Senseman *et al*., 2014).

Environmentally, glyphosate binds tightly to organic matter in soils and has a relatively short half-life due to microbial degradation, reducing concerns of soil residual activity and movement into groundwater (Duke and Powles, 2008). Though the EPA and several researchers have found very low mammalian toxicity with glyphosate, the International Agency for Research on Cancer has reclassified it as a possible carcinogen (Williams *et al*., 2000; Duke and Powles, 2008; IARC, 2017; US EPA, 2023). However, the recent publicity surrounding the safety of glyphosate has led some counties and municipalities in Florida to locally ban its use on public lands and waters (Thompson, 2019). Research by De María *et al*. (2021) found that glyphosate and amino methylphosphonic acid (AMPA), a chemical formed from the metabolism of glyphosate, were commonly present in several waterbodies of south Florida, leading to repeated detection of both compounds at extremely low levels in the blood plasma of Florida manatees (*Trichechus manatus latirostris*).

Metsulfuron-methyl is one of a few current alternatives to glyphosate for *L. microphyllum* management (Hutchinson *et al*., 2006). Metsulfuron-methyl is a sulfonylurea herbicide that translocates symplastically through the phloem and works by inhibiting branched chain amino acid production in plants through the inhibition of the acetolactate synthase (ALS) enzyme (Senseman *et al*., 2014). The herbicide is absorbed rapidly through leaves, accumulating in the actively growing meristematic tissue. Similar to glyphosate, symptoms slowly appear over one to four weeks and include chlorosis of new leaves, followed by whole plant yellowing and shoot necrosis. However, growth immediately ceases within hours (Senseman *et al*., 2014).

Metsulfuron-methyl is relatively short lived in the environment, but its fate is dependent upon several factors including soil pH, organic matter, and microbial activity (Walker *et al*., 1989). When present in the soil, root uptake is also possible, warranting some non-target concerns (Kotoula-Syka *et al*., 1993; Boutin *et al*., 2000). Metsulfuron-methyl is practically non-toxic to mammals, is rapidly

Table 1. Herbicides used in Florida, USA for control of *Lygodium microphyllum*.

Herbicide active ingredient	WSSA mechanism of action and (group)	Typical concentration range	Selectivity
Glyphosate	EPSP synthase inhibition (9)	7.2–14.4 g L^{-1}	Non-selective
Metsulfuron	Acetolactate synthase inhibition (2)	0.15 g L^{-1}	Selectively controls some dicots and monocots, but safe on many grasses
Triclopyr	Auxin mimic (4)	5.4–10.8 g L^{-1}	Selectively controls most dicots and certain monocots, safe on grasses
Florpyrauxifen-benzyl	Auxin mimic (4)	0.08–0.16 g L^{-1}	Selectively controls many monocots and some dicots but safe on many grasses

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excreted in urine and does not bioaccumulate. However, common native wetland ferns such as swamp fern (*Telmatoblechnum serrulatum),* chain fern (*Woodwardia virginica),* hottentot fern *(Thelypteris interrupta),* and cinnamon fern *(Osmundastrum cinnamomea),* are highly susceptible to metsulfuron-methyl (Taylor, 2006; Hutchinson and Langeland, 2008).

The final widely used herbicide for *L. microphyllum* control is triclopyr. Triclopyr is a pyridine carboxylic acid that mimics the plant hormone, auxin, or indole acetic acid (IAA). The mode of action results in inhibition of meristems at high concentrations and unregulated cell division and growth at low concentrations (Senseman *et al*., 2014; Heap, 2025). Triclopyr is absorbed through foliage and translocates symplastically to actively growing tissues including *L. microphyllum* rhizomes (Hutchinson *et al*., 2010). Leaf curling and cupping, and stem bending, twisting, and swelling all appear within a few days of application (Senseman *et al*., 2014). Triclopyr selectively controls many dicots but can safely be used over many monocots (primarily Poaceae) without injury (Lewer and Owen, 1990).

Due to issues with non-target damage, potential herbicide resistance, and the need for longer-term *L. microphyllum* control, additional chemistries are needed to increase the available tools. In general, new aquatic herbicide registrations are infrequent and warrant testing on *L. microphyllum* to determine if they may be useful. Florpyrauxifen-benzyl demonstrates systemic activity and selective control at low use rates on aquatic weeds, including Eurasian watermilfoil (*Myriophyllum spicatum*) and hydrilla (*Hydrilla verticillata*) (Netherland and Richardson, 2016; Richardson *et al*., 2016; Beets *et al*., 2019; Sperry *et al*., 2021). Early testing indicated selectivity to desirable native plant species, warranting testing with *L. microphyllum* (Netherland and Richardson, 2016). In subsequent small plot testing and operational research, Glueckert *et al*. (2023) found florpyrauxifen-benzyl resulted in a slow defoliation of *L. microphyllum* over 60 to 90 days. Control was generally maintained for 6–13 months, but recovery was evident by 17 months after treatment. Non-target data was not a component of these initial efficacy studies, and warrants further testing.

Two new triclopyr formulations and florpyrauxifen-benzyl are the only new herbicides recently tested for *L. microphyllum* control. Glueckert *et al*. (2023) compared the new triclopyr acid and choline formulations to the commonly used triethylamine formulation and the glyphosate commercial standard. They found the acid and choline formulations were comparable in activity to the triethylamine formulation and the glyphosate commercial standard at both short (6 months after treatment) and long-term (28 months after treatment) evaluations. However, the acid formulation was applied at a 30% lower triclopyr concentration than the choline or triethylamine formulations. This represents a significant reduction in herbicide into the wetland environment. This reduction in herbicide applied compared to other treatments has stimulated considerable interest for controlling *L. microphyllum* (Bell *et al*., 2023).

Although limited to a minimal selection of herbicides, new application techniques and technology may make the application of these herbicides more effective at controlling the fern. While *L. microphyllum* occurs widespread throughout wetland ecosystems, populations are strongly associated with hydrology and certain native plant communities (Fujisaki *et al*., 2010). Recent efforts include optimized monitoring and detection of the fern so that land managers may prioritize and visit more infestations. Bonneau *et al.* (2019) utilized aerial imagery obtained by Unmanned Aerial Systems (i.e., drones, UAS) and satellites to apply a model improving detection of the fern, potentially increasing the number of controlled sites by 34% annually compared to non-computational observation-based detection and monitoring.

Integrating remote sensing is important in both plant detection and mapping, and the emergence of interest in autonomous application of herbicides by UAS. Precision agriculture has driven the innovation of UAS applied herbicides, which is now being adopted in natural areas. Takekawa *et al*. (2023) used a spray UAS in managed wetlands in California and found that the integration of UAS for both mapping and spraying perennial pepperweed (*Lepidium latifolium*) provided effective control and reduced overall costs. Future innovations to herbicide application must continue to account for non-target impact on sensitive wetland environments.

PHYSICAL, CULTURAL, AND BIOLOGICAL CONTROL AND INTEGRATED MANAGEMENT

Physical removal of *L. microphyllum* is often only used when site-specific conditions dictate it and is often integrated with either biological control and/or chemical control (Hutchinson *et al*., 2006). As mentioned above, fire often occurs naturally in these systems, and is frequently used as a cultural management tool. Stocker *et al*. (2008) found that when fire and herbicide were integrated to treat *L. microphyllum*, native plant diversity and frequency increased and species richness recovered to pre-treatment levels within 2 years. This, however, hinged on the biannual treatment of *L. microphyllum* with herbicides. Richards *et al*. (2020) reported that *L. microphyllum* suffered damage to the hydraulic function of the rhizome, and recovered significantly slower than plants that simply sustained mechanical damage. The findings in Richards *et al*. (2020) appear to be site specific though. When David *et al*. (2020) conducted a similar *in-situ* experiment in three different sites within the greater Everglades ecosystem (including Everglades National Park), they found that *L. microphyllum* recovered to pre-fire coverage within 5 months. Both fire studies show clearly that fire is highly integratable with herbicides and biological control. Season may contribute to differences in burn survival of *L. microphyllum* plants observed between the works of Richards *et al*. (2020) and David *et al*. (2020), as was observed following burns of Cape Sable infestations (Sebesta, 2021). Individual *L. microphyllum* plants, surveyed and tagged pre-burn, were monitored for regrowth for 2 years following prescribed burns in Everglades National Park. Some were subjected to a dry season burn, while others a wet season burn. Both season burns reduced *L. microphyllum* biomass and eliminated small plants completely; however, mortality was greater following the dry season burn, as was the reduction in reproductive output (sporulation delayed for 9 months versus6 months following the wet season burn) (Sebesta, 2021).

Biological control for *L. microphyllum* is more than 30 years in development (Pemberton, 1998). Though this amount of time often eclipses careers, it is often just barely the amount of time needed to discover, verify safety through host range testing, and begin to see impacts (Hinz *et al*., 2019). To date, three agents have been released in Florida: 2 moths in the Crambidae (Lepidoptera), *Austromusotima camptozonale* (Hampson), and *Neomusotima conspurcatalis* (Warren), and the mite, *Floracarus perrepae* Knihinicki & Boczek (Acariformes: Eriophyidae). Approximately 50,000 *A. camptozonale* were released in Florida, but this agent failed to establish (Boughton and Pemberton, 2008). *F. perrepae* and *N. conspurcatalis* are both established and widely distributed, although both present certain challenges in regards to distribution. *F. perrepae* triggers the formation of galls on the apical meristems and leaf margins which results in a reduction of rachis length and biomass (David and Lake 2020; Goolsby *et al*., 2005) (Fig. 5). *F. perrepae* exhibits hyperspecificity, only galling about half of the plants in Florida (Goolsby *et al*., 2005).

N. conspurcatalis is widely distributed throughout the *L. microphyllum* range, but displays “boom and bust” population cycles (Jones and Lake, 2020; Smith *et al*., 2014). Jones and Lake (2020) showed that herbivory from *N. conspurcatalis* is capable of changing plant architecture (e.g., more lateral branching, less climbing), and inducing mortality in young plants when they are environmentally stressed (e.g., shade). Despite its ephemeral nature, *N. conspurcatalis* was found to be established at 69% of the establishment survey sites (Smith *et al*., 2014) (Fig. 6).

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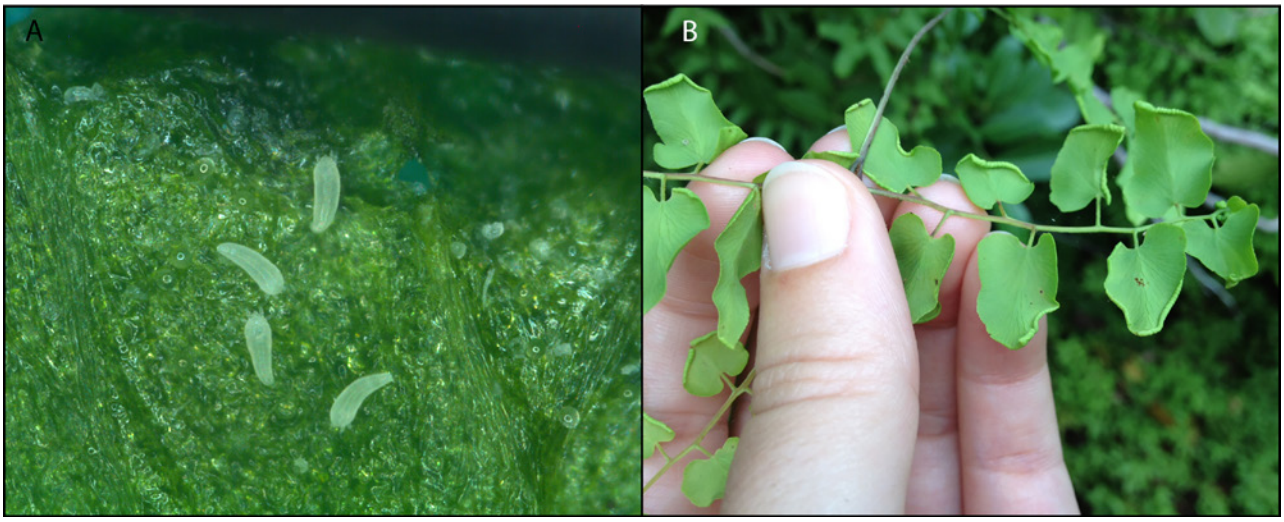


Fig. 5. *Floracarus perrepae* (A) adult mites (photo credit: E.C. Lake), and (B) galling on pinnae margins (photo credit: M.C. Smith).



Fig. 6. *Neomusotima conspurcatalis* (A) larvae (multiple instars), (B) adult (photo credit: M.C. Smith), and (C) damage from large outbreak on an Everglades tree island (photo credit: C. Mason).

A combination of biological, cultural, mechanical, and chemical methods is often needed for effective *L. microphyllum* management in Florida to protect native plant communities and natural ecosystems. Land managers agree that integrated methods along with adequate funding, extensive monitoring, and cooperation among stakeholders on public and private land is the only way to achieve adequate control of *L. microphyllum* on a landscape level (Hutchinson and Langeland, 2006). Preliminary research on management approaches that incorporate biological control, herbicide treatments, and prescribed fire, have shown that an integrated approach has the potential to be an effective strategy for *L. microphyllum* (Stocker *et al*., 2008; Hutchinson and Langeland, 2010; David *et al*., 2020, 2021).

In some cases, biological control on its own may take decades to achieve success, requiring herbicide treatments to supplement them (Ainsworth, 2003). Studies involving the use of herbicides in the presence of biological control agents are intrinsically complex because chemical treatments may be toxic to the agents, who are also dependent on sufficient populations of the host plant to maintain their own population (Ainsworth, 2003). On Everglades tree islands treated with herbicides, the *Lygodium* gall mite, *F. perrepae,* likely experiences extirpation after herbicide

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treatments (David *et al*., 2021). David *et al*. (2021) determined that the mite recolonized in large numbers and preferentially attacked young sporophytes on islands recovering from herbicide treatment, while nearly absent on untreated islands. The paucity of the mite on untreated island, though, could be the consequence of hard conditions for searching (e.g., high and dense vegetation). The mite also exhibited a preference for tree islands treated with glyphosate rather than triclopyr (David *et al*., 2021). While the cause of this preference is unknown, these results suggest that biological control is compatible with herbicides, but the choice of herbicide may impact recolonization (David *et al*., 2021).

Prescribed fire alone will not suppress *L. microphyllum* but integrating fire with herbicide and/or biological control agents can be a management tactic. Prescribed fire burns thick rachis mats and reduces above ground biomass, improving access to the site for herbicide applicators when re-growth begins (Stocker *et al*., 2008). In prescribed fire/biannual herbicide plots, Stocker *et al*. (2008) reduced herbicide use by 50% compared to reapplications in herbicide only plots. *L. microphyllum* biological control agents also respond positively to post-fire regrowth. Following prescribed burns on Cape Sable, mite populations re-established on recovered *L. microphyllum* plants within 9 months, with damage levels (% galling of frond biomass) surpassing those observed for undamaged fronds (Sebesta, 2021). *Neomusotima conspurcatalis* also naturally colonizes *L. microphyllum* regrowth following prescribed burns (David *et al*., 2020).

Biotic resistance and climate factors may limit establishment and impact of biologial control agents (Schwarzländer *et al*., 2018). The larval stages of both moths are susceptible to predation by ants (Jones *et al*., 2021). For example, *Pseudomyrmex gracilis* Fabricus individuals have been observed predating on *N. conspurcatalis* larvae during biological control releases, (EC Lake, pers obs) and *A. comptozonale* in field sites (Boughton and Pemberton, 2008; Rayamajhi *et al*., 2014). *N. conspurcatalis* is attacked by 13 known parasitoids, and has likely accumulated more (Boughton *et al*., 2012; Lake *et al*., 2015, 2019). Predatory ants are likely attracted to the fern by its foliar nectar production (Sebesta *et al*., 2018). Site hydrology may also affect *L. microphyllum* survival of prescribed burning and ant densities. This is the case for a biological control agent used on *Senecio jacobaea, Tyria jacobaea* (Crider, 2011). Predation on *T. jacobaea* by *Camponotus* ants varied with hydrological conditions of the sites and it is not unreasonable to suspect a similar dynamic in Everglades habitats (Crider, 2011).

For all forms of control, seasonality appears to play a major role, due to its strong influence on growth and reproduction of *L. microphyllum* (David *et al*., 2020). For herbicides, their activity in the soil is decreased drastically under hydric conditions, whereas this is the peak growing season for *L. microphyllum* (Hutchinson, 2010; Senseman *et al*., 2014). Biological control agents also display this mismatch. Their populations exhibit two peaks around fall and late spring and populations decrease during summer, possibly due to reaching upper thermal thresholds (Allen *et al*., 2014; David *et al*., 2020).

The future of *Lygodium microphyllum* research

GENOMIC INSIGHTS

Genomic studies of invasive plants provide valuable insights into how species colonize and rapidly spread in novel environments. For plants, this is partly attributted to various ecological mechanisms and genetic factors (Maron *et al*., 2004; Whitney and Gabler, 2008; Gioria *et al*., 2023). Analyses of the genomics of populations from both source and invaded ranges can reveal patterns of phylogeography and historical demography, and the roles of evolutionary processes, such as genetic drift, gene flow, mutation, and selection (Welles and Dlugosch, 2019). Although invasives are expected to experience severe genetic bottlenecks during establishment (i.e., the "genetic paradox of invasion", Baker and Stebbins, 1965), recent work has shown that such decreases are rarely substantial enough to impact invasion success and may be ameliorated by multiple introductions, admixture, and *de novo* mutation (Dlugosch and Parker, 2008). The type of variation retained during colonization, rather than the total amount, may largely dictate invasion success and the role of genetics in invaders (Dlugosch *et al*., 2015). Recent studies have shown that assessing neutral genetic variation is not generally consistent with adaptive variation (e.g., Leinonen *et al*., 2008). Our point here focuses on the idea that the genetic architecture of adaptive variation (and therefore traits that may be important to invasion success) is an essential consideration, rather than the total amount of genetic variation. As Dlugosch *et al*. (2015) state: "many studies of introduced species indicate that genetic might play an integral role in the progress of invasion, but they suggest that role is determined by *what* genetic variation is introduced, rather than *how much.*" For example, loci of large effect and structural variants appear to play a prominent role in adaptation in novel environments. Loci of large effect are more likely to suffer from bottlenecks, whereas traits that are controlled by many loci of small effect should experience less of an impact of founder events. Understanding how the underlying genetics control traits relevant to invasion success is an essential aspect of invasion genomics.

Sequencing projects targeting ferns have greatly expanded the genomic resources for this clade. A recent international consortium has sequenced the nuclear genome of *L. microphyllum* (Pelosi *et al*., 2025), detailing fundamental differences in the biology between gametophyte and sporophyte life phases, such as differential gene expression and molecular strategies in response to freezing. The genomic traits of this species are largely similar to other ferns, and the development of these data has the potential to facilitate the generation of more effective and efficient management strategies using a genome-informed approach. For example, using a genome-wide genotyping by sequencing (GBS) method has been adopted to determine the route of invasion of *L. microphyllum* in Florida and assess a possible link between genetics and biological control establishment (D. Brooks, *pers comm.*). These lineages likely correspond to galling and non-galling populations. This may (and even likely) represents a host and mite host range mismatch despite previous analyses showing convergence (McCulloch *et al*., 2018). These data emphasize that there is likely a genetic component to the establishment of the mite biological control, and a recently awarded United States Department of Agriculture (USDA) funded study aims to determine the genomic and molecular underpinnings of mite establishment using functional genomic approaches. Further research is warranted to conduct analysis on the genetic diversity of *L. microphyllum* across its invasive range in Florida. Such research is crucial to assess the susceptibilty and potenital for adapatition to control measures, including the devlopment of resistance to herbicide or biological control agents.

NEW BIOLOGICAL CONTROL AGENTS

Three additional agents beyond the two established agents have completed their host range testing and are currently in the permitting and approval pipeline at the USDA Animal Plant Health Inspection Service (USDA, 2024). These include a relatively large moth, *Callopistria exotica* Guenee (Lepidoptera: Noctuidae), a sawfly *Neostrombocerus albicomus* (Konow) (Hymenoptera: Tenthridinidae), and another moth in the Crambidae, *Lygomusotima stria* Solis & Yen (Solis *et al*., 2004; Madeira *et al*., 2016; Wheeler *et al*., 2024, 2025). Anecdotally, the larvae of *C. exotica* and *N. albicomus* exhibit significant defoliation in laboratory rearing (G. Wheeler and E.C. Lake, *pers comm.*). These additional agents were pursued due to their potential to cause consistent significant defoliation including population outbreaks. For example, several species of sawfly experience large population eruptions that can be very damaging to their host (e.g., *Tsuga, Pinus, Fagus* species, among others) (Taniwaki *et al*., 2022). Additionally, *C. exotica* shows the ability to utilize *L. microphyllum* and the invasive

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congenere, *L. japonicum* equally well, increasing the potential for its impact beyond Florida (Wheeler *et al*., 2024). Although these defoliating species may compete with *N. conspurcatalis,* they have the potential to be much more consistently damaging throughout the range than the brown Lygodium moth, and perhaps be subject to fewer impacts from parasitism, though this cannot be fully studied until their release (Lake *et al*., 2019).

L. microphyllum continues to spread in Florida northward and into remote areas of conservation land (e.g., Everglades National Park, Big Cypress National Preserve, Fakahatchee Strand State Park, Big Cypress Seminole Indian Reservation) despite significant amounts of time and resources devoted to understanding how *L. microphyllum* establishes and spreads, and what, if any, vulnerabilities may exist. Like, other emblematic invasive species in Florida, *Melaleuca quinquenervia* Cav. Blake (Myrtaceae) for example, reducing the spread and impact of *L. microphyllum* will require more research and resources devoted to understanding and testing IPM strategies (Smith, 2022). That is, refining the application of biological, chemical, and cultural treatment techniques in addition to understanding the underlying mechanisms of genetics, plant chemistry and the herbivore/plant interactions. This project also highlights the significant contribution of multi-agency and multi-national partnerships – although we are far from the endpoint with *L. microphyllum* management*,* we have far more answers and a much clearer direction in which to take this research than we did 20 years ago.

CONFLICT OF INTEREST

The authors declare that there are no competing interests.

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References

Allen, J.L., Clusella-Trullas, S. and Chown, S.L. (2014) Thermal tolerance of *Cyrtobagous salviniae*: A biocontrol agent in a changing world. *BioControl* 59, 357–366. DOI: 10.1007/s10526-014-9570-2.

Baker, H.G. and Stebbins, G.L. (1965) The genetics of colonizing species. In: *Proceedings of the First International Union of Biological Sciences Symposia on General Biology*. Academic Press, New York, p. 588.

Barrett, M.A., Brandt, L.A. and Thomas, B. (2006) Monitoring ground treatments of Old World climbing fern (*Lygodium microphyllum*) on the Arthur R. Marshall Loxahatchee NWR: A follow up report. *Wildland Weeds* 10, 4–7.

Beckner, J. (1968) *Lygodium microphyllum*, another fern escaped in Florida. *American Fern Journal* 58, 93–94. DOI: 10.2307/1546255.

Beets, J., Heilman, M. and Netherland, M.D. (2019) Large-scale mesocosm evaluation of florpyrauxifen-benzyl, a novel arylpicolinate herbicide, on Eurasian and hybrid watermilfoil and seven native submersed plants. *Journal of Aquatic Plant Management* 57, 49–55.

Bell, M.G., Enloe, S.F., Leary, J.J.K. and Lauer, D. (2023) A contractor comparison of novel IPT tools and techniques for Brazilian peppertree (*Schinus terebinthifolia*) management. *Invasive Plant Science and Management* 16, 170–176. DOI: 10.1017/inp.2023.22.

Bonneau, M., Martin, J., Peyrard, N., Rodgers, L., Romagosa, C.M. *et al.* (2019) Optimal spatial allocation of control effort to manage invasives in the face of imperfect detection and misclassification. *Ecological Modelling* 392, 108–116. DOI: 10.1016/j.ecolmodel.2018.11.012.

Boughton, A.J. and Pemberton, R.W. (2008) Efforts to establish a foliage-feeding moth, *Austromusotima camptozonale*, against *Lygodium microphyllum* in Florida, considered in the light of a retrospective review of establishment success of weed biocontrol agents belonging to different arthropod taxa. *Biological Control* 47, 28–36. DOI: 10.1016/j.biocontrol.2008.07.007.

Boughton, A.J., Kula, R.R., Gates, M., Zhang, Y., Nunez, M. *et al.* (2012) Parasitoids attacking larvae of a recently introduced weed biological control agent, *Neomusotima conspurcatalis* (Lepidoptera: Crambidae): Key to species, natural history, and integrative taxonomy. *Annals of the Entomological Society of America* 105, 753–767. DOI: 10.1603/AN11157.

Boutin, C., Lee, H.-B., Peart, E.T., Batchelor, P.S. and Maguire, R.J. (2000) Effects of the sulfonylurea herbicide metsulfuron methyl on growth and reproduction of five wetland and terrestrial plant species. *Environmental Toxicology and Chemistry* 19, 2532–2541. DOI: 10.1002/etc.5620191020.

Brandt, L.A. and Black, D.W. (2001) Impacts of the Introduced Fern, *Lygodium Microphyllum*, on the Native Vegetation of Tree Islands in the Arthur R. Marshall Loxahatchee National Wildlife Refuge. *Florida Scientist* 64, 191–196.

Brandt, L.A., Ecker, D., Rivera, I.G., Traut, A. and Mazzotti, F.J. (2003) Wildlife and vegetation of bayhead islands in the ARM Loxahatchee National Wildlife Refuge. *Southeastern Naturalist* 2, 179–194. DOI: 10.1656/1528-7092(2003)002[0179:WAVOBI]2.0.CO;2.

Call, E.M., Brandt, L.A. and DeAngelis, D.L. (2007) Old world climbing fern (*Lygodium microphyllum*) spore germination in natural substrates. *Florida Scientist* 70, 55–61.

Crider, K.K. (2011) Predator interference with the cinnabar moth (*Tyria jacobeae*) for the biological control of tansy ragwort (*Senecia jacobaea*). *Invasive Plant Science and Management* 4, 332–340. DOI: 10.1614/IPSM-D-10-00082.1

David, A.S. and Lake, E.C. (2020) Eriophyid mite *Floracarus perrepae* reduces climbing ability of the invasive vine *Lygodium microphyllum*. *Biological Control* 146, 104271. DOI: 10.1016/j.biocontrol.2020.104271.

David, A.S., Sebesta, N., Abdel-Kader, A.A. and Lake, E.C. (2020) Colonization by biological control agents on post-fire regrowth of invasive *Lygodium microphyllum* (Lygodiaceae). *Environmental Entomology* 49, 796–802. DOI: 10.1093/ee/nvaa076.

David, A.S., Glueckert, J.S., Enloe, S.F., Carmona, C.A., Abdel-Kader, A.A. *et al.* (2021) Eriophyid mite *Floracarus perrepae* readily colonizes recovering invasive vine *Lygodium microphyllum* following herbicide treatment. *BioControl* 66, 573–584. DOI: 10.1007/s10526-021-10087-6.

De María, M., Silva-Sanchez, C., Kroll, K.J., Walsh, M.T., Nouri, M.-Z. *et al.* (2021) Chronic exposure to glyphosate in Florida manatee. *Environment International* 152, 106493. DOI: 10.1016/j.envint.2021.106493.

Dlugosch, K.M. and Parker, I.M. (2008) Founding events in species invasions: Genetic variation, adaptive evolution, and the role of multiple introductions. *Molecular Ecology* 17, 431–449. DOI: 10.1111/j.1365-294X.2007.03538.x.

Dlugosch, K.M., Anderson, S.R., Braasch, J., Cang, F.A. and Gillette, H.D. (2015) The devil is in the details: Genetic variation in introduced populations and its contributions to invasion. *Molecular Ecology* 24, 2095–2111. DOI: 10.1111/mec.13183.

Duever, M.J., Meeder, J.F., Meeder, L.C. and McCollum, J.M. (1994) The climate of south florida and its role in shaping the everglades ecosystem.

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In: Davis, S. , Ogden, J.C. , Park, W.A. (eds) *Everglades: The Ecosystem and its Restoration*. CRC Press, Boca Raton, FL, pp. 225–248.

Duke, S.O. and Powles, S.B. (2008) Glyphosate: A once-in-a-century herbicide. *Pest* 64, 319–325. DOI: 10.1002/ps.1518.

Fujisaki, I., Brandt, L.A., Chen, H. and Mazzotti, F.J. (2010) Colonization, spread, and growth of *Lygodium microphyllum* on tree islands in a wetland in Florida. *Invasive Plant Science and Management* 3, 412–420. DOI: 10.1614/IPSM-D-10-00023.1.

Getsinger, K.D., Netherland, M.D. and Stubbs, D.R. (2011) Aquatic herbicide registration new model. In: Leslie, A.R. and Westbrooks, R.G. (eds) *Invasive Plant Management Issues and Challenges in the United States: 2011 Overview*. American Chemical Society Symposium Series, Washington, DC, pp. 45–50.

Gioria, M., Hulme, P.E., Richardson, D.M. and Pyšek, P. (2023) Why are invasive plants successful? *Annual Review of Plant Biology* 74, 635–670. DOI: 10.1146/annurev-arplant-070522-071021.

Glueckert, J.S., Leary, J.J.K. and Enloe, S.F. (2023) Evaluation of novel triclopyr formulations for control of Old World climbing fern (*Lygodium microphyllum*). *Invasive Plant Science and Management* 16, 73–80. DOI: 10.1017/inp.2023.6.

Goolsby, J.A., Zonneveld, R., Makinson, J.R. and Pemberton, R.W. (2005) Host-range and cold temperature tolerance of *Floracarus perrepae* Knihinicki & Boczek (Acari: Eriophyidae), a potential biological-control agent of *Lygodium microphyllum* (Pteridophyta: Lygodiaceae). *Australian Journal of Entomology* 44, 321–330. DOI: 10.1111/j.1440-6055.2005.00470.x.

Gordon, D.R. (1998) Effects of invasive, non-indigenous plant species on ecosystem processes: Lessons from Florida. *Ecological Applications* 8, 975–989. DOI: 10.1890/1051-0761(1998)008[0975:EOINIP]2.0.CO;2.

Gunderson, L. and Snyder, J.R. (1997) Fire patterns in the southern everglades. In: Davis, S.M. and Ogden, J.C. (eds) *Everglades: The Ecosystem and Its Restoration*. St. Lucie Press, Boca Raton, FL, pp. 291–305.

Hanks, J.G. (1998) A Monographic Study of *Lygodium* Swartz (Pteridophyta: Lygodiaceae). Doctor of Philosophy Dissertation, City University of New York. Available at: https://www.proquest.com/openview/4fa3c073088fd5b74479991709da1053/1?cbl=18750&diss=y&pq-origsite=gscholar

Heap, I. (2025) The International Herbicide-Resistant Weed Database. Available at: https://www.weedscience.org/ (accessed 20 February 2025).

Hiatt, D., Serbesoff-King, K., Lieurance, D., Gordon, D.R. and Flory, S.L. (2019) Allocation of invasive plant management expenditures for conservation: Lessons from Florida, USA. *Conservation Science and Practive* e51, 1–10. DOI: 10.1111/csp2.51.

Hinz, H.L., Winston, R.L. and Schwarzländer, M. (2019) How safe is weed biological control? A global review of direct nontarget attack. *The Quarterly Review of Biology* 94, 1–27. DOI: 10.1086/702340.

Hutchinson, J.T. (2010) Physiological characteristics of herbicides and management of Old World climbing fern (*Lygodium microphyllum*). Dpctpr of Philosophy Dissertation, University of Florida. 124pp.

Hutchinson, K.A.L. and Hutchinson, J. (2013) Natural Area Weeds: Old World Climbing Fern (*Lygodium microphyllum*). Available at: http://edis.ifas.ufl.edu/ag122 (accessed 16 September 2013).

Hutchinson, J.T. and Langeland, K.A. (2006) Survey of control measures on old world climbing fern (*Lygodium microphyllum*) in Southern Florida. *Florida Scientist* 69, 217–223.

Hutchinson, J.T. and Langeland, K.A. (2007) *Final Report: Evaluation of Current and Potential New Herbicides to Control Lygodium microphyllum Florida Department of Environmental Protection*. Bureau of Invasive Plant Management, Tallahassee, FL.

Hutchinson, J.T. and Langeland, K.A. (2008) Response of selected nontarget native Florida wetland plant species to metsulfuron methyl. *Journal of Aquatic Plant Management* 46, 72–76.

Hutchinson, J.T. and Langeland, K.A. (2010) Review of two non-native, invasive climbing ferns (*Lygodium japonicum* and *L. microphyllum*), sympatric records and additional distribution records from Florida. *American Fern Journal* 100, 57–66. DOI: 10.1614/IPSM-D-13-00074.1.

Hutchinson, J.T. and Langeland, K.A. (2012) Repeated herbicide application for control of old world climbing fern (*Lygodium microphyllum*) and the effects on nontarget vegetation on everglade tree islands. *Invasive Plant Science and Management* 5, 477–486. DOI: 10.1614/IPSM-D-12-00015.1.

Hutchinson, J.T. and Langeland, K.A. (2014) Tolerance of *Lygodium microphyllum* and *L. japonicum* spores and gametophytes to freezing temperature. *Invasive Plant Science and Management* 7, 328–335.

Hutchinson, J.T., Ferriter, A.P., Serbesoff-King, K., Langeland, K.A. and Rodgers, L. (2006) *Old World Climbing Lygodium Management Plan for Florida Exotic Pest Plant Council Lygodium Task Force*. West Palm Beach, FL, USA.

Hutchinson, J.T., Langeland, K.A., MacDonald, G.E. and Querns, R. (2010) Absorption and translocation of glyphosate, metsulfuron, and triclopyr in old world climbing fern (*Lygodium microphyllum*). *Weed Science* 58, 118–125. DOI: 10.1614/WS-D-09-00046.1.

IARC (2017) *IARC Monographs on the Evaluation of Carcinogenic Risks to Humans—Volume 112: Some Organophosphate Insecticides and Herbicides International Agency for Research on Cancer*. Lyon, France.

Jones, I.M. and Lake, E.C. (2020) Defoliation of the invasive fern *Lygodium microphyllum* by *Neomusotima conspurcatalis*: Effects on plant performance across a range of light conditions. *Biological Control* 144, 104236. DOI: 10.1016/j.biocontrol.2020.104236.

Jones, I.M., Madeira, P.T., Blair, J.Z. and Lake, E.C. (2021) Using molecular gut content analysis to identify key predators in a classical weed biological control system: A study with *Neomusotime conspurcatalis* (Lepidoptera: Crambidae). *Biocontrol* 66, 825–836. DOI: 10.1007/s10526-021-10090-x.

Kotoula-Syka, E., Eleftherohorinos, I.G., Gagianas, A.A. and Sficas, A.G. (1993) Phytotoxicity and persistence of chlorsulfuron, metsulfuron-methyl, triasulfuron and tribenuron-methyl in three soils. *Weed Research* 33, 355–367. DOI: 10.1111/j.1365-3180.1993.tb01951.x.

Lake, E.C., Gates, M.W., Smith, M.C., Witkus, G.L. and Pratt, P.D. (2015) First report of an egg parasitoid reared from *Neomusotima conspurcatalis* (Lepidoptera: Crambidae), a biological control agent of *Lygodium microphyllum* (Schizaeales: Lygodiaceae). *Florida Entomologist* 98, 1244–1246. DOI: 10.1653/024.098.0436.

Lake, E.C., Kula, R.R., Gates, M.W., Smith, M.C., Minteer, C.R. *et al.* (2019) The first pupal parasitoids of neomusotima conspurcatalis warren (Lepidoptera: Crambidae), a biological control agent of *Lygodium microphyllum* (Cav.) R. Br. (Polypodiales: Lygodiaceae) in Florida. *Proceedings of the Entomological Society of Washington* 121, 314–319. DOI: 10.4289/0013-8797.121.2.314.

Leinonen, T., O'hara, R.B., Cano, J.M. and Merilä, J. (2008) Comparative studies of quantitative trait and neutral marker divergence: A meta-analysis. *Journal of Evolutionary Biology* 21, 1–17. DOI: 10.1111/j.1420-9101.2007.01445.x.

Lewer, P. and Owen, W.J. (1990) Selective action of the herbicide triclopyr. *Pesticide Biochemistry and Physiology* 36, 187–200. DOI: 10.1016/0048-3575(90)90010-Y.

Lockwood, J.L., Ross, M.S. and Sah, J.P. (2003) Smoke on the water: The interplay of fire and water flow on Everglades restoration. *Frontiers in Ecology and the Environment* 1, 462–468. DOI: 10.1890/1540-9295(2003)001[0462:SOTWTI]2.0.CO;2.

Lott, M.S., Volin, J.C., Pemberton, R.W. and Austin, D.F. (2003) The reproductive biology of the invasive ferns *Lygodium microphyllum* and *L. japonicum* (Schizaeaceae): Implications for invasive potential. *American Journal of Botany* 90, 1144–1152. DOI: 10.3732/ajb.90.8.1144.

Loveless, C.M. (1959) A study of the vegetation in the Florida Everglades. *Ecology* 40, 1–9. DOI: 10.2307/1929916.

Lynch, R.L., Chen, H., Brandt, L.A. and Mazzotti, F.J. (2009) Old World climbing fern (*Lygodium microphyllum*) invasion in hurricane-caused treefalls. *Natural Areas Journal* 29, 210–215. DOI: 10.3375/043.029.0302.

Madeira, P.T., Facey, J., Pratt, P.D., Maul, D.P. and Wheeler, G. (2016) Are three colonies of *Neostromboceros albicomus*, a candidate biological control agent for *Lygodium microphyllum*, the same host biotype? *Biocontrol Science and Technology* 26, 440–445. DOI: 10.1080/09583157.2015.1123676.

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Maron, J.L., Vilà, M., Bommarco, R., Elmendorf, S. and Beardsley, P. (2004) Rapid evolution of an invasive plant. *Ecological Monographs* 74, 261–280. DOI: 10.1890/03-4027.

McCulloch, G.A., Makinson, J.R., Zonneveld, R., Purcell, M.F., Raghu, S. *et al.* (2018) Assessment of genetic structuring in the Lygodium fern moths *Austromusotima camptozonale* and *Neomusotima conspurcatalis* in their native range: Implications for biological control. *Biological Control* 121, 8–13. DOI: 10.1016/j.biocontrol.2018.01.012.

Munns, R. and Termaat, A. (1986) Whole-plant responses to salinity. *Functional Plant Biology* 13, 143–160. DOI: 10.1071/pp9860143.

Netherland, M.D. and Richardson, R.J. (2016) Evaluating sensitivity of five aquatic plants to a novel arylpicolinate herbicide utilizing an organization for economic cooperation and development protocol. *Weed Science* 64, 181–190. DOI: 10.1614/WS-D-15-00092.1.

Norris, R.F., Caswell-Chen, E.P. and Kogan, M. (2003) *Concepts in Integrated Pest Management Prentice Hall*. Upper Saddle River, New Jersey, USA.

Pelosi, J.A., Zumwalde, B.A., Testo, W.L., Kim, E.H., Burleigh, J.G. *et al.* (2024) All tangled up: Unraveling phylogenetics and reticulate evolution in the vining ferns, Lygodium (Schizaeales). *American Journal of Botany* 111, 1–21. DOI: 10.1002/ajb2.16389.

Pelosi, J.A., Sorojsrisom, E.S., Barbazuk, W.B. and Sessa, E.B. (2025) Population genomics of the gametophyte-only fern *Vittaria appalachiana* provides insights into clonal plant evolution. *New Phytologist* 246, 349–364. DOI: 10.1111/nph.20433.

Pemberton, R.W. (1998) The potential of biological control to manage old world climbing fern (*Lygodium microphyllum*), an invasive weed in Florida. *American Fern Journal* 88, 176. DOI: 10.2307/1547770.

Pemberton, R.W. (2003) Northward range extension in florida of the invasive fern *Lygodium Microphyllum* (lygodiaceae). *SIDA, Contributions to Botany* 20, 1759–1761.

Pemberton, R.W. and Ferriter, A.P. (1998) Old world climbing fern (*Lygodium microphyllum*), a dangerous invasive weed in Florida. *American Fern Journal* 88, 165–175. DOI: 10.2307/1547769.

Pemberton, R.W., Goolsby, J.A. and Wright, T. (2002) Old world climbing fern. In: *Biological Control of Invasive Plants in the Eastern United States, USDA Forest Service Publication*. USDA Forest Service Publication, FHTET, pp. 139–147.

Rayamajhi, M.B., Pratt, P.D., Leidi, J.G. and Center, T.D. (2014) *Austromusotima camptozonale* (Lepidoptera: Crambidae) herbivory results in frond and rhizome mortality of the invasive fern *Lygodium microphyllum* (Schizaeales: Lygodiaceae). *Florida Entomologist* 97, 1308–1316.

Richards, J.H., Sebesta, N. and Taylor, J. (2020) Fire effects on growth of the invasive exotic fern *Lygodium microphyllum* and implications for management. *Management of Biological Invasions* 11, 541–559.

Richardson, R.J., Haug, E. and Netherland, M.D. (2016) Response of seven aquatic plants to a new arylpicolinate herbicide. *Journal of Aquatic Plant Management* 54, 26–31.

Roberts, D. (1996) Climbing fern wreaks wetland havoc. *Florida Department of Environmental Protection Resource Management Notes* 8, 1–13.

Rodgers, L., Pernas, T. and Hill, S.D. (2014) Mapping invasive plant distributions in the Florida everglades using the digital aerial sketch mapping technique. *Invasive Plant Science and Management* 7, 360–374. DOI: 10.1614/IPSM-D-12-00092.1.

Schwarzländer, M., Hinz, H.L., Winston, R.L. and Day, M.D. (2018) Biological control of weeds: An analysis of introductions, rates of establishment and estimates of success, worldwide. *BioControl* 63, 319–331. DOI: 10.1007/s10526-018-9890-8.

Sebesta, N. (2021) Invasion ecology and response to fire of the nonnative fern *Lygodium microphyllum* in the South Florida Everglades. Doctor of Philosophy Dissertation, Florida International University. Available at: https://digitalcommons.fiu.edu/etd/4820/

Sebesta, N., Richards, J. and Taylor, J. (2016) The effects of heat on spore viability of *Lygodium microphyllum* and implications for fire management. *Southeastern Naturalist* 15, 40–50. DOI: 10.1656/058.015.sp804.

Sebesta, N., Jones, I.M. and Lake, E.C. (2018) First report of foliar nectar production by *Lygodium microphyllum*, an invasive fern in Florida. *American Fern Journal* 108, 180–183. DOI: 10.1640/0002-8444-108.4.180.

Senseman, S.A., Hancock, H.G., Wauchope, R.D., Armhurst, K.L., Peters, T.J. *et al.* (2014) *Herbicide Handbook, 10th edn*. Herbicide Handbook Committee, Weed Science Society of America, Lawrence, KS. 513pp.

Smith, M.C. (2022) Successful melaleuca biological control in the Florida Everglades. In: Van Driesche, R.G. , Winston, R.L. , Perring, T.M. , Lopez, V.M. (eds) *Contributions of Classical Biological Control to the U.S. Food Security, Forestry, and Biodiversity. FHAAST-2019-05.* USDA Forest Service, Morgantown, West Virginia, pp. 356–367.

Smith, M.C., Lake, E.C., Pratt, P.D., Boughton, A.J. and Pemberton, R.W. (2014) Current status of the biological control agent *Neomusotima conspurcatalis* (Lepidoptera: Crambidae), on *Lygodium microphyllum* (Polypodiales: Lygodiaceae) in Florida. *Florida Entomologist* 97, 817–820.

Solis, M.A., Yen, S.-H. and Goolsby, J.H. (2004) Species of *Lygomusotima* new genus and *Neomusotima* Yoshiyasu (Lepidoptera: Crambidae) from Australia and Southeastern Asia Feeding on *Lygodium microphyllum* (Schizaeaceae). *Annals of the Entomological Society of America* 97, 64–76. DOI: 10.1603/0013-8746(2004)097[0064:SOLNGA]2.0.CO;2.

Soti, P.G., Jayachandran, K., Purcell, M., Volin, J.C. and Kitajima, K. (2014) Mycorrhizal symbiosis and *Lygodium microphyllum* invasion in South Florida – A biogeographic comparison. *Symbiosis* 62, 81–90. DOI: 10.1007/s13199-014-0272-4.

Sperry, B.P., Leary, J.K., Jones, K.D. and Ferrell, J.A. (2021) Observations of a submersed field application of florpyrauxifen-benzyl suppressing hydrilla in a small lake in central Florida. *Journal of Aquatic Plant Management* 59, 20–26.

Stocker, R.K., Ferriter, A.P., Thayer, D.D., Rock, M. and Smith, S. (1997) Old World climbing fern hitting south Florida below the belt. *Wildland Weeds* 1, 6–11.

Stocker, R.K., Miller, R.E., Black, D.W., Ferriter, A.P. and Thayer, D.D. (2008) Using fire and herbicide to control *Lygodium microphyllum* and effects on a pine flatwoods plant community in South Florida. *Natural Areas Journal* 28, 144–154. DOI: 10.3375/0885-8608(2008)28[144:UFAHTC]2.0.CO;2.

Takekawa, J.Y., Hagani, J.S., Edmunds, T.J., Collins, J.M., Chappell, S.C. *et al.* (2023) The sky is not the limit: Use of a spray drone for the precise application of herbicide and control of an invasive plant in managed wetlands. *Remote Sensing* 15, 3845. DOI: 10.3390/rs15153845.

Taniwaki, T., Watanabe, K., Komine, H., Tochigi, K., Yamane, M. and Koike, S. (2022) Response of specialist and generalist predators to nonprogressive annual fluctuations in herbivorous insect populations. *Biological Control* 165, 104810. DOI: 10.1016/j.biocontrol.2021.104810.

Taylor, J. (2006) Remote infestations of *Lygodium microphyllum*: A case study at Everglades National Park. *Wildland Weeds* 9, 28–29.

Thomas, B. and Brandt, L.A. (2003) Monitoring ground treatments of Old World climbing fern (*Lygodium microphyllum*) on the Arthur R. Marshall Loxahatchee NWR. *Wildland Weeds* 6, 9–11.

Thompson, B. (2019) FWC to Resume Improved Aquatic Plant Spray Program When the Time Comes: County Spraying Still on Pause. Available at: https://www.chronicleonline.com/news/local/fwc-to-resume-improved-aquatic-plant-spray-program-when-the-time-comes-county-spraying-still/article_ca7e3324-36d9-11e9-afab-7f9e2f5962e9.html (accessed 2 April 2021).

US EPA (2023) EPA Posts Final Endangered Species Act Biological Opinion for Enlist Herbicide Products. Available at: https://www.epa.gov/pesticides/epa-posts-final-endangered-species-act-biological-opinion-enlist-herbicide-products (accessed 30 May 2024).

USDA (2024) Technical Advisory Group for Biological Control Agents of Weeds Manual U.S. Dept. of Agriculture, Washington, D.C. USA, Interim Edition.

Volin, J.C., Lott, M.S., Muss, J.D. and Owen, D. (2004) Predicting rapid invasion of the Florida Everglades by Old World Climbing Fern (*Lygodium microphyllum*). *Diversity and Distributions* 10, 439–446. DOI: 10.1111/j.1366-9516.2004.00091.x.

Volin, J.C., Kruger, E.L., Volin, V.C., Tobin, M.F. and Kitajima, K. (2010) Does release from natural belowground enemies help explain the invasiveness of *Lygodium microphyllum*? A cross-continental comparison. *Plant Ecology* 208, 223–234. DOI: 10.1007/s11258-009-9700-6.

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Walker, A., Cotterill, E.G. and Welch, S.J. (1989) Adsorption and degradation of chlorsulfuron and metsulfuron-methyl in soils from different depths. *Weed Research* 29, 281–287. DOI: 10.1111/j.1365-3180.1989.tb00913.x.

Welles, S.R. and Dlugosch, K.M. (2019) Population genomics of colonization and invasion. In: Rajora, O.P. (ed) *Population Genomics: Concepts, Approaches and Applications*. Springer International Publishing, Cham, pp. 655–683.

Wheeler, G.S., Lake, E.C., Mattison, E. and Sutton, G.F. (2024) Host range, biology, and climate suitability of *Callopistria exotica*, a potential biological control agent of Old World climbing fern (*Lygodium microphyllum*) in the USA. *Biological Control* 188, 105410. DOI: 10.1016/j.biocontrol.2023.105410.

Wheeler, G.S., Lake, E.C., Mattison Frank, E., Pratt, P.D. and Sutton, G. (2025) Host range, biology, and thermal tolerance of *Lygomusotima stria*, a potential biological control agent of old world climbing fern (*Lygodium microphyllum*) in the USA. *Biological Control* 204, 105741. DOI: 10.2139/ssrn.5106447.

Whitney, K.D. and Gabler, C.A. (2008) Rapid evolution in introduced species, 'invasive traits' and recipient communities: Challenges for predicting invasive potential. *Diversity & Distributions* 14, 569–580. DOI: 10.1111/j.1472-4642.2008.00473.x.

Williams, G.M., Kroes, R. and Munro, I.C. (2000) Safety evaluation and risk assessment of the herbicide roundup and its active ingredient, glyphosate, for humans. *Regulatory Toxicology and Pharmacology* 31, 117–165. DOI: 10.1006/rtph.1999.1371.

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